TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	76.6 ± 3.20
Correlation Stability	89.2 ± 1.40
Distribution Consistency	84.5 ± 2.42
Entropy Preservation	87.4 ± 1.74
Mean Z Score Drift	0.2 ± 0.04
Categorical Fidelity	83.0 ± 0.45
Privacy Risk	49.2 ± 1.54
Data Density	100.0 ± 0.01
Coverage Score	20.0 ± 6.07
Feature Accuracy	77.2 ± 3.46
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	41.483	-3.263	7.3%	OK
TB	3.299	1.368	-1.931	58.5%	Drift
DB	1.486	0.515	-0.971	65.4%	Drift
Alkphos	290.576	202.865	-87.712	30.2%	Drift
Sgpt	80.714	35.521	-45.192	56.0%	Drift
Sgot	109.911	40.476	-69.434	63.2%	Drift
TP	6.483	6.743	+0.260	4.0%	OK
ALB	3.142	3.395	+0.253	8.0%	OK
A/G Ratio	0.947	0.976	+0.029	3.1%	OK
Selector	1.286	1.466	+0.179	13.9%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	76.6 ± 3.20
Correlation Stability (*2)	89.2 ± 1.40
Distribution Consistency (*3)	84.5 ± 2.42
Entropy Preservation (*4)	87.4 ± 1.74
Mean Z Score Drift (*5)	0.2 ± 0.04
Categorical Fidelity (*6)	83.0 ± 0.45
Privacy Risk (*7)	49.2 ± 1.54
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	20.0 ± 6.07

Feature Accuracy (*10)	77.2 ± 3.46
Novelty Score (*11)	0.4 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.5581196581196581	0.6965811965811965	-0.006837606837606836	
F1	0.8108734553482932	0.6432352726469053	0.795882766893805	-0.014990688454488255	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6786324786324787	0.5470085470085471	0.688034188034188	0.009401709401709413	
F1	0.7865029989624185	0.6352910592060004	0.7905153983660848	0.004012399403666234	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6598290598290599	0.52991452991453	0.6521367521367522	-0.007692307692307676	
F1	0.7625770352986383	0.6270137006653225	0.7561042563880139	-0.006472778910624377	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5726495726495726	0.6803418803418804	0.005982905982905984	
F1	0.7821343746683586	0.6610651230037546	0.7848524376018178	0.002718062933459186 5	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6136752136752137	0.6914529914529914	0.005128205128205132 5	
F1	0.7971299319768493	0.7298891575980087	0.8010446057348431	0.003914673757993481	

Model: XGB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6837606837606838	0.5512820512820513	0.6940170940170943	0.010256410256410253	
F1	0.7935686358268239	0.6458486185214328	0.8026568444387188	0.009088208611895015	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7111111111111111	0.5777777777777777	0.6957264957264957	-0.015384615384615396	
F1	0.8138214323544078	0.6698288682547469	0.7986879996640673	-0.015133432690340643	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6717948717948719	0.7068376068376068	0	
F1	0.8279091624529537	0.7921929365209561	0.8279091624529537	0	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5786324786324786	0.5752136752136752	0.016239316239316272	
F1	0.5587432119380917	0.6427939536717413	0.5869274975342524	0.028184285596160707	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Male	0.8	0.1	146	17	29	5.9	3.2	1.18
2	17	Female	0.7	0.2	145	18	36	7.2	3.9	1.18
3	42	Male	8.9	4.5	272	31	61	5.8	2.0	0.5
4	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
5	17	Male	0.9	0.2	224	36	45	6.9	4.2	1.55
6	34	Female	0.8	0.2	192	15	12	8.6	4.7	1.2
7	29	Female	1.2	0.4	150	25	30	6.5	3.1	0.85
8	54	Male	0.9	0.3	200	40	50	7.0	3.5	1.0
9	38	Female	2.5	1.0	180	35	45	6.8	3.4	1.02
10	45	Male	3.0	1.5	220	50	60	7.2	3.8	1.1
11	60	Female	1.8	0.9	175	30	40	6.2	3.0	0.95
12	22	Male	0.6	0.2	160	28	35	6.7	3.2	1.1
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	22	Male	0.9	0.3	179	18	21	6.7	3.7	1.2
2	66	Female	2.9	1.3	168	21	38	5.5	1.8	0.4
3	45	Male	2.3	1.3	282	132	368	7.3	4.0	1.2
4	72	Male	0.6	0.1	102	31	35	6.3	3.2	1.0
5	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
6	60	Male	0.5	0.1	500	20	34	5.9	1.6	0.37
7	29	Female	1.2	0.5	210	35	40	6.5	3.2	1.0
8	54	Male	0.9	0.3	180	22	30	7.0	3.5	0.9
9	39	Female	2.1	0.9	250	45	55	7.2	4.0	1.1
10	47	Male	1.5	0.7	300	50	60	6.8	3.8	0.8
11	34	Female	0.6	0.2	160	28	35	5.9	2.8	1.2
12	62	Male	2.5	1.0	400	75	80	7.5	4.2	0.7
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Female	2.6	1.2	410	59	57	5.6	3.0	0.8
2	28	Female	1.0	0.3	90	18	108	6.8	3.1	0.8
3	29	Male	0.7	0.2	165	55	87	7.5	4.6	1.58
4	49	Male	2.0	0.6	209	48	32	5.7	3.0	1.1
5	74	Female	0.9	0.3	234	16	19	7.9	4.0	1.0
6	23	Female	2.3	0.8	509	28	44	6.9	2.9	0.7
7	34	Female	1.2	0.5	210	45	30	6.2	3.5	1.3
8	29	Male	0.9	0.3	150	22	25	7.0	3.0	1.0
9	42	Female	2.5	1.0	300	50	40	6.5	3.8	1.2
10	55	Male	0.7	0.2	180	30	35	5.8	2.9	1.1
11	60	Female	1.5	0.6	250	55	45	6.0	3.2	1.0
12	38	Male	2.1	0.8	220	40	50	6.4	3.6	1.1
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	18	Male	0.7	0.1	312	308	405	6.9	3.7	1.1
2	60	Male	5.7	2.8	214	412	850	7.3	3.2	0.78
3	58	Female	2.4	1.1	915	60	142	4.7	1.8	0.6
4	38	Female	2.6	1.2	410	59	57	5.6	3.0	0.8
5	64	Male	1.4	0.5	298	31	83	7.2	2.6	0.5
6	57	Male	4.0	1.9	190	45	111	5.2	1.5	0.4
7	22	Female	1.0	0.3	150	30	40	6.5	3.0	0.8
8	37	Male	3.2	1.5	200	50	60	7.0	3.5	1.1
9	45	Female	2.5	1.0	300	45	55	6.8	3.2	0.9
10	29	Male	1.5	0.5	180	40	50	7.2	4.0	1.3
11	54	Female	0.9	0.2	220	20	30	5.5	2.8	0.7
12	33	Male	1.1	0.4	160	35	45	6.2	3.1	0.95
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
2	78	Male	1.0	0.3	152	28	70	6.3	3.1	0.9
3	32	Male	0.9	0.3	462	70	82	6.2	3.1	1.0
4	50	Male	1.2	0.4	282	36	32	7.2	3.9	1.1
5	34	Male	5.9	2.5	290	45	233	5.6	2.7	0.9
6	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
7	29	Female	0.9	0.3	210	45	50	6.5	3.2	0.8
8	54	Male	1.5	0.5	300	90	100	7.0	3.5	1.1
9	37	Female	0.8	0.2	180	28	35	6.2	3.0	1.0
10	45	Male	2.1	0.8	220	60	70	6.8	3.4	1.2
11	50	Female	1.0	0.3	290	40	45	6.6	3.1	0.9
12	62	Male	0.7	0.1	150	20	30	5.9	2.8	0.93
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	39	Male	1.6	0.8	230	88	74	8.0	4.0	1.0
2	60	Male	8.9	4.0	950	33	32	6.8	3.1	0.8
3	30	Female	0.8	0.2	198	30	58	5.2	2.8	1.1
4	24	Male	1.0	0.2	189	52	31	8.0	4.8	1.5
5	18	Male	1.4	0.6	215	440	850	5.0	1.9	0.6
6	41	Male	0.9	0.2	169	22	18	6.1	3.0	0.9
7	25	Female	1.2	0.4	180	25	30	6.5	3.5	1.0
8	34	Male	2.5	1.0	250	40	45	7.0	3.0	1.2
9	29	Female	0.9	0.3	200	35	50	5.8	2.9	0.9
10	52	Male	3.8	1.5	310	60	70	6.2	3.1	1.1
11	41	Female	1.5	0.6	220	30	25	7.5	4.0	1.2
12	38	Male	0.7	0.2	160	20	15	6.0	2.5	0.8
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	39	Male	6.6	3.0	215	190	950	4.0	1.7	0.7
2	30	Male	0.8	0.2	174	21	47	4.6	2.3	1.0
3	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
4	57	Male	4.0	1.9	190	45	111	5.2	1.5	0.4
5	52	Male	0.9	0.2	156	35	44	4.9	2.9	1.4
6	35	Male	1.6	0.7	157	15	44	5.2	2.5	0.9
7	34	Female	1.2	0.5	150	30	45	6.0	3.0	1.0
8	29	Male	0.8	0.3	180	20	25	5.5	2.8	0.9
9	45	Male	2.5	1.0	200	50	60	6.5	3.2	1.1
10	52	Female	3.0	1.2	250	40	55	7.0	3.5	1.2
11	38	Male	1.5	0.6	220	35	50	6.8	3.4	1.0
12	41	Female	0.9	0.2	190	25	30	5.9	2.9	0.8
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Female	2.6	1.2	410	59	57	5.6	3.0	0.8
2	48	Male	1.6	1.0	588	74	113	7.3	2.4	0.4
3	75	Male	2.9	1.3	218	33	37	3.0	1.5	1.0
4	23	Male	1.0	0.3	212	41	80	6.2	3.1	1.0
5	60	Male	1.5	0.6	360	230	298	4.5	2.0	0.8
6	38	Female	0.8	0.2	145	19	23	6.1	3.1	1.03
7	29	Female	1.2	0.5	210	45	50	6.0	3.2	0.9
8	34	Male	0.9	0.3	150	30	40	5.5	2.8	1.1
9	45	Female	2.5	1.0	300	60	70	7.0	3.5	0.8
10	52	Male	1.8	0.7	250	50	65	6.5	3.0	0.9
11	39	Female	0.6	0.2	180	25	30	5.8	2.9	1.0
12	60	Male	3.0	1.5	400	70	80	8.0	3.8	0.7
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	1.0	0.3	230	48	58	8.4	4.2	1.0
2	38	Female	0.8	0.2	145	19	23	6.1	3.1	1.03
3	41	Female	0.9	0.2	201	31	24	7.6	3.8	1.0
4	47	Male	0.9	0.2	265	40	28	8.0	4.0	1.0
5	51	Male	0.8	0.2	230	24	46	6.5	3.1	nan
6	32	Male	25.0	13.7	560	41	88	7.9	2.5	2.5
7	34	Female	1.1	0.4	210	30	25	6.5	3.2	0.8
8	45	Male	0.9	0.3	150	22	18	7.0	3.5	1.0
9	29	Female	0.8	0.2	180	28	20	5.9	2.9	1.1
10	52	Male	1.5	0.7	300	40	35	6.8	3.4	0.9
11	60	Female	0.6	0.1	190	25	15	5.5	2.5	1.0
12	37	Male	2.1	1.0	250	50	45	7.2	3.8	1.2
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	72	Male	0.6	0.1	102	31	35	6.3	3.2	1.0
2	65	Male	4.9	2.7	190	33	71	7.1	2.9	0.7
3	37	Male	1.8	0.8	215	53	58	6.4	3.8	1.4
4	36	Male	5.3	2.3	145	32	92	5.1	2.6	1.0
5	56	Male	17.7	8.8	239	43	185	5.6	2.4	0.7
6	42	Female	0.9	0.2	165	26	29	8.5	4.4	1.0
7	29	Female	1.2	0.4	210	22	28	6.5	3.0	1.1
8	54	Male	2.5	1.1	180	45	50	7.2	3.5	1.4
9	39	Female	0.9	0.3	220	30	35	5.8	2.9	0.9
10	45	Male	3.1	1.5	250	60	70	6.8	3.2	1.0
11	62	Female	1.4	0.6	190	25	30	6.0	3.1	1.0
12	34	Male	0.8	0.2	175	20	25	7.0	3.8	1.2
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3